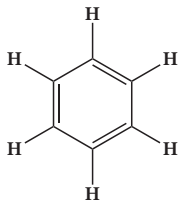
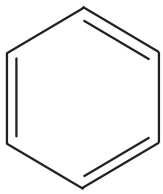
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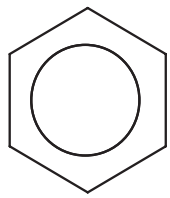
8. Representation of benzene:

Hence, there are three ways in which benzene can be represented.



Expanded form


Kekule structure


Short hand representation

Benzene and its homologous series

Benzene and its homologous series are colorless liquids with pleasant odour .They are lighter than water and insoluble in it. Their vapours are highly flammable, and volatile and toxic in nature.

13.5.4. Sources of aromatic compound:

- Benzene and other aromatic compound are obtained from coal tar and petroleum
- It can also be prepared in laboratory using some simple aliphatic compounds

1. Preparation of benzene

(i) industrial preparation of benzene from coal tar :

Coal tar is a viscous liquid obtained by the pyrolysis of coal. During fractional distillation, coal tar is heated and distills away its volatile compounds namely benzene, toluene, xylene in the temperature range of 350 to 443 K. These vapours are collected at the upper part of the fractionating column (Table 13.5.)

TABLE.13.5COMPENENTS OF DISTILLATION OF COAL TAR

NAME OF THE FRACTION	TEMPERATURE RANGE	NAME OF THE COMPENENTS
1. Crude light oil	350 - 443 K	Benzene, Toluene, Xylenes
2. Middle oil	443 - 503 K	Phenol, Naphthalene
3. Heavy oil	503 - 543 K	Naphthalene, Cresol
4. Green oil	543 - 633 K	Anthracene
5. Pitch	Alone 633 K	Residue

(ii) from acetylene.

Acetylene on passing through a red –hot tube trimerises to give benzene. We have